AIE1901: Large Language Models for Social Simulations

Final Project

School of Artificial Intelligence, The Chinese University of Hong Kong, Shenzhen

Deliverables

Submit: A single ZIP file named:
`AIE1901_Final_GroupX.zip`

Inside ZIP:
- `slides.pdf` or `slides.pptx`
- source code, and data
- `README.txt` (a simple instruction for code implementations)
- `video.mp4` (a short 3-minute video)

Short Video ($\leq$ 3 minutes)

In addition to the slides, produce a short video summary of your project, similar to your mid-term video. This can take one of the following forms:
- **Mini Presentation:** A concise overview of your LLM-based simulation, covering the key ideas from your slides.
- **Edited Summary Video:** A short video combining simulation outputs, visualizations, and brief narration to highlight your system and findings.

Requirements:
- Length: $\leq$ 3 minutes.
- Must contain substantive content related to your project.
- Should summarize what you built, how it works, and what results you obtained.

Purpose

The final project focuses on your understanding, implementation, and creative designs of LLM-driven social simulations. This is an open-ended project emphasizing creativity, correctness, and clarity of communication. You have the following two options to choose from.

Option A.
- Successfully run the provided Lab 5 notebook on LLM-based simulations.
- Implement two or more variations or extensions from the suggestion list in the notebook, **or** design your own ideas.

Option B.
- Alternatively, you may choose to ignore Lab 5 entirely and **build your own LLM-based simulation from scratch** and run your code successfully with visualizations.

Content Requirements for Slides

Your slides should follow the structure below and present the content clearly with visuals and concise explanations.

1. Overview

- State the objective of your project.
- If you used the Lab 5 notebook, briefly describe its baseline design.
- If you built your own simulation, introduce the scenario or application.

2. System Design & Methodology

- Describe the architecture of your code.
- Explain agent roles, prompting strategies, and environment setup.
- Include diagrams or flowcharts.
- If you built upon Lab 5, clearly state what you changed. Highlight modifications, extensions, or custom modules in your code.

3. Experiments & Results

- Describe your experimental setup and scenarios.
- Present results such as plots, tables, or agent interaction transcripts.
- Analyze how different parameters or prompt strategies affect behaviors.
- Compare variants if applicable.

4. Insights, Challenges, and Reflections

- Summarize major insights about LLM-based simulation design.
- Discuss unexpected behaviors or technical challenges.
- Reflect on limitations and future improvements.

5. Group Member Contributions

- Summarize major contributions of each group member if a team of 2 or more people is formed.

Grouping

Students may:

- freely self-organize into groups, or
- choose to present individually.

Assessment Rubric (100 pts)

Criterion	Description	Points
Technical Correctness	Whether your simulation runs properly; correctness of implementation	25

Creativity & Extension	Depth and originality of variations or self-designed system	25
Experimental Design	Quality of experiments and clarity of results	20
Presentation Quality	Structure, clarity, and visuals in your slides	20
Professionalism	Formatting, naming, and completeness of submission (including video)	10